

Proporzioni e Algoritmi

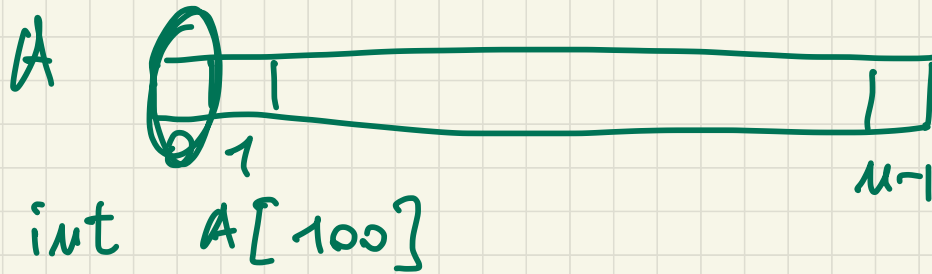
10/10/22



INSERTION SORT

Input: array A di n numeri interi.

Output: array A ORDINATO: $A[1] \leq A[2] \leq \dots \leq A[n]$



$$|A| = n$$

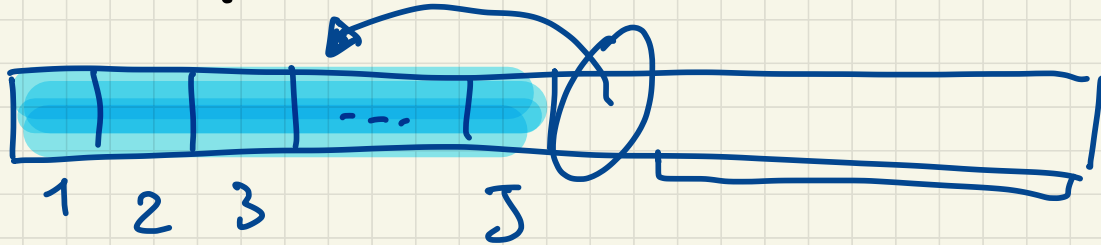
INSERTION SORT: soluzione "incrementale",
con un metodo iterativo

INSERTION-SORT procede in modo ITERATIVO

Al passo j^{esimo}

$\forall j = 2, \dots, n$

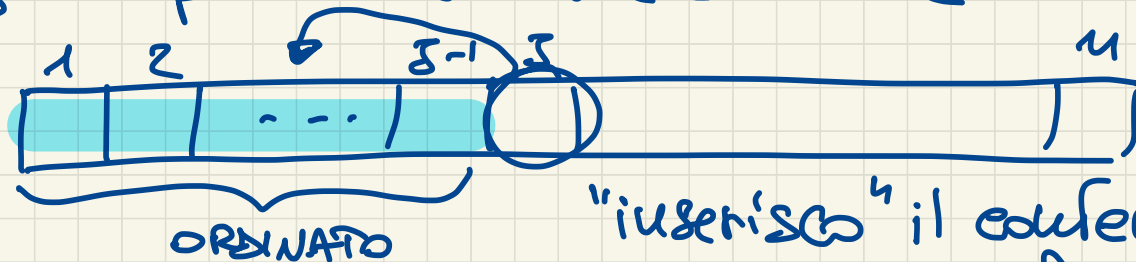
- l'elemento di indice j dell'array viene inserito al posto giusto tra i primi j elementi
- al termine del passo j^{esimo} , i primi j elementi sono dunque ordinati



ORDINATO

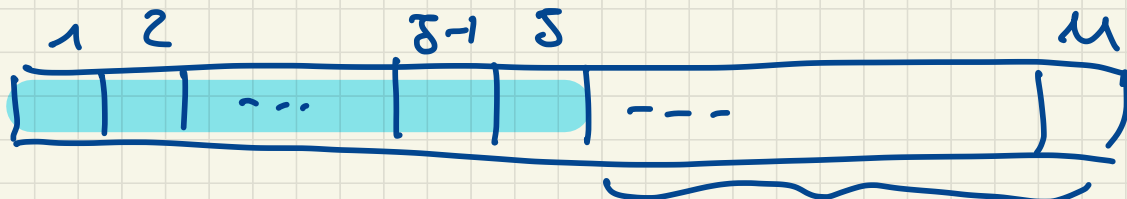
$\forall j = 2 \rightarrow n$ ($n-1$ iterazioni)

- al primo passo ho inizialmente



"inserisco" il contenuto di $A[j]$
al posto giusto tra i primi
 $j+1$ elementi

- e alla fine ho



gli elementi ancora
da ordinare

ESEMPIO

A

5	2	4	6	1	3
1	2	3	4	5	6

$n=6$

$J=2$

2	5	4	6	1	3
---	---	---	---	---	---

insertisco $A[2]=2$

$J=3$

2	5	4	6	1	3
---	---	---	---	---	---

insertisco $A[3]=4$

2	4	5	6	1	3
---	---	---	---	---	---

$J=4$

2	4	5	6	1	3
---	---	---	---	---	---

insertisco $A[4]=6$

$J=5$

2	4	5	6	1	3
---	---	---	---	---	---

insertisco $A[5]=1$

1	2	4	5	6	3
---	---	---	---	---	---

$J=6=n$

1	2	4	5	6	3
---	---	---	---	---	---

insertisco $A[6]=3$

1	2	3	4	5	6
---	---	---	---	---	---

PSEUDOCODE

INSERTION SORT (A)

$$|A| = n$$

FOR $j = 2$ TO n DO

INDICAMENTE $i++$

$\{ k = A[j];$
 $i = j - 1;$

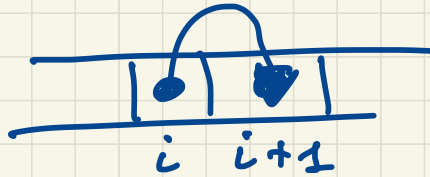
WHILE $(i > 0 \text{ OR } A[i] > k)$ DO

$\{ A[i+1] = A[i]$

$i--;$

$A[i+1] = k$

CERCA LA
POSIZIONE
IN CUI
INSERIRE k



$i = i - 1$

	1	2	3	4	5	6
1	5	2	4	6	1	3
2	2	5	4	6	1	3
3	2	4	5	6	1	3
4	1	2	4	5	6	3
5	1	2	4	5	6	3

$$j = 2 \quad k = 2 \Rightarrow A[2] = 5$$

$$j = 3 \quad k = 4 \Rightarrow A[3] = 5 \quad A[1] = k = 2$$

$$j = 4 \quad k = 6 \Rightarrow A[4] = k = 6$$

$$j = 5 \quad k = 1$$

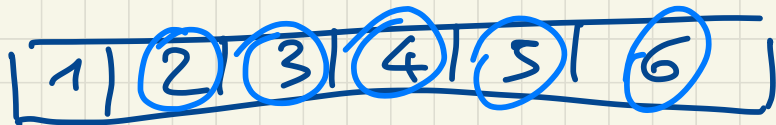
... OMISSE

INSERTION SORT (A) $ A =n$	COSTO	# VOLTE ESEGUITO
<u>FOR</u> $j=2$ <u>TO</u> n <u>DO</u> -----	COSTANTE: $\Theta(1)$	$n-1$
$\{ k = A[j];$ -----	$\Theta(1)$	$n-1$
$i = j-1;$ -----	$\Theta(1)$	$n-1$
<u>WHILE</u> $(i > 0 \text{ \&\& } A[i] > k)$ $\Theta(1)$	$\Theta(1)$	$\sum_{j=2}^n t_j$
$\{ A[i+1] = A[i];$ -----	$\Theta(1)$	$\sum_{j=2}^n (t_j - 1)$
$i--;$ -----	$\Theta(1)$	$\sum_{j=2}^n (t_j - 1)$
$\{ A[i+1] = k;$ -----	$\Theta(1)$	$n-1$

t_j : # VOLTE CHE SI VAUTA LA GUARDA DEL
WHILE NELLA j ESIMA ITERAZIONE DEL FOR

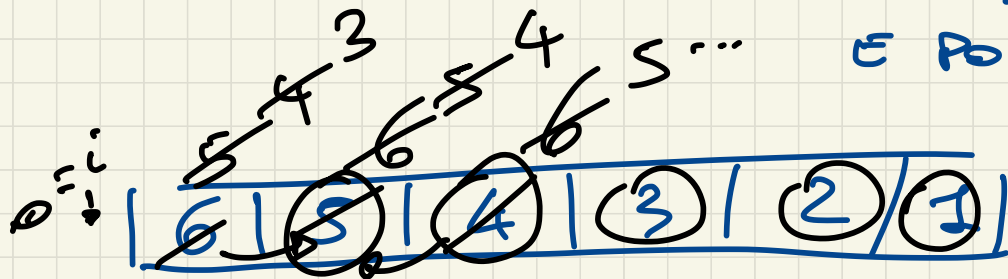
$$T(n) = C(n-1) + C(n-1) + C(n-1) + C \sum_{j=2}^n t_j + C \sum_{j=2}^n (t_j - 1) + C \sum_{j=2}^n (t_j - 1) + C(n-1)$$

CASO OTTIMO $\rightarrow \forall j \quad L_j = 1$
 OVVERO $\forall j \quad A[i] \leq k$



A più ordinato

CASO PESSIMO $\rightarrow \forall j \quad L_j = j$
 OVVERO PER $j-1$ ELEMENTI
 HO $A[i] > k$
 E POI $L = \emptyset$



A ordinato in
 senso inverso

$$T(n) = C(n-1) + C(n-1) + C(n-1) + C \sum_{j=2}^n t_j + C \sum_{j=2}^n (t_j - 1) + C \sum_{j=2}^n (t_j - 1) + C(n-1)$$

↳ USE OPTIM
($t_j = 1 \forall j$)

$$\Rightarrow T(n) = 4C(n-1) + C \sum_{j=2}^n 1 + 2 \cdot C \sum_{j=1}^{n-1} 1 = 2C(1+1+\dots+1) = C(1+1+\dots+1) = C(n-1)$$

$$= 5C(n-1) + 2C(n-1) =$$

$$= 7C(n-1) = \cancel{7Cn} - \cancel{7C} \in \Theta(n)$$

INSERTION SORT $\in$ LINEARE AL USE
OPTIMO!

$$T(n) = \cancel{C(n-1)} + \cancel{C(n-1)} + \cancel{C(n-1)} + C \sum_{j=2}^n t_j +$$

$$C \sum_{j=2}^n (t_j - 1) + C \sum_{j=2}^n (t_j - 1) + \cancel{C(n-1)}$$

↳ CASO PESSIMO

$$(t_j = j \quad \forall j)$$

INSERTION SORT
QUADRÁTICO AL
CASO PESSIMO

$$= 4C(n-1) +$$

$$C\left(\frac{n^2 + n}{2} - 1\right) +$$

$$2C\left(\frac{n^2 - n}{2}\right)$$

$$T(n) = 4C(n-1) + C \sum_{j=2}^n j +$$

$$+ 2 \cdot C \cdot \sum_{j=2}^n (j-1) =$$

$$2C(1+2+\dots+n-1)$$

$$= 2C\left(\frac{n(n-1)}{2}\right)$$

$$C(2+3+4+\dots+n)$$

$$= C\left(\frac{n(n+1)}{2} - 1\right)$$

$$= 4Cn - 4C + C\frac{n^2}{2} + \frac{Cn}{2} - C + Cn^2 - Cn$$

$$= \left(C \cdot \frac{3}{2}\right)n^2 + \left(4C + \frac{C}{2} - C\right)n - 5C \in \Theta(n^2)$$

4 INVARIANTE di CICLO di INSERTION SORT

- proprietà di A all'uscita della j -esima iterazione del for

$$\forall j = 2 \dots n$$

chi è ordinato?
e dove stanno quelli ordinati?